

DEEPAK CHAPAGAIN

Software Engineer specializing in React Native, Native Android Integration & Applied AI

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PROFESSIONAL SUMMARY

Software engineer with 6+ years shipping production React and React Native applications, specializing in native Android integration (Kotlin, TurboModules) and applied AI systems (RAG, embeddings). Delivered measurable business impact, including a 22% lift in checkout approval rates, a 40% cut in API over-fetching, and a hands-free communication platform deployed to 2,500+ retail stores. WCAG 2.1 AA-focused, CI/CD-fluent, and React-certified (Infosys).

TECHNICAL SKILLS

Frontend & Mobile: React, React Native (New Architecture, TurboModules), TypeScript/JavaScript, Redux/Redux-Saga, React Router/Navigation

Native & Systems Integration: Kotlin bridges, Bluetooth/audio pipelines, Android foreground services, WebView + JWT authentication

APIs, Testing & DevOps: REST/GraphQL, Jest/React Testing Library/Vitest, GitHub Actions, Jenkins, Git, SonarQube

Applied AI & Observability: RAG, embeddings, OpenAI API, Dynatrace, Quantum Metric, Splunk, Crashlytics

Accessibility & Cloud: WCAG 2.1 AA, WAVE, Microsoft Azure (AZ-900 certified)

PROFESSIONAL EXPERIENCE

Tractor Supply Company (via Infosys Limited), Technology Consultant I

Sept 2024 – Present

Brentwood, TN (Hybrid)

TSC Connect – React Native, Kotlin, Bluetooth, TurboModules

- **Built** a hands-free Android communication app live across **2,500+** Tractor Supply stores, powering a real-time push-to-talk communication platform for store associates.
- Bridged TypeScript and Kotlin via React Native's New Architecture/TurboModules to integrate wired headset controls, text-to-speech, and foreground services for reliable background operation.
- Implemented an NLP voice-command engine (phonetic matching, similarity scoring, state-aware sessions) that reduced false triggers and improved hands-free reliability.

CMA (Consumer Mobile Application) – React Native, GraphQL, WebView, JWT

- Integrated a fraud-detection vendor into mobile checkout, cutting login errors **~10%**, lifting conversion **1.45 pts**, and improving approval rate **22%**.
- Led the REST-to-GraphQL migration, reducing over-fetching **~40%** and improving p90 render time by **~120ms**.
- Implemented secure WebView/JWT-to-cookie session sync and built Dynatrace, Quantum Metric, and Splunk monitoring to strengthen reliability and speed up diagnosis.

Southern California Edison (via Infosys Limited), Technology Analyst

Nov 2020 – Aug 2024

Richardson, TX (Remote)

- Delivered end-to-end enterprise web features (React, Redux, TypeScript, REST) for SCE's customer-facing platform.
- Refactored legacy class components to React Hooks and unified state management with Redux-Saga, cutting bundle size **~9%** and improving development velocity.
- Built WCAG 2.1 AA-aligned UI components, achieving a **75%+** WAVE accessibility pass rate.
- Established Jest/React Testing Library practices, raising unit test coverage to **80%** overall and **85%** on pull requests.
- Built Dynatrace, Quantum Metric, and Splunk dashboards for proactive monitoring, cutting incident reports **~12%**.
- Drove Agile delivery, including sprint planning, code review, deployment support, and production issue resolution.

Allstate Corporation (via Infosys Limited), Senior Associate

Feb 2020 – Aug 2020

Irving, TX

- Integrated and tested SiteCatalyst analytics middleware within native mobile apps, ensuring accurate data collection for business reporting.
- Validated microservice API endpoints for mobile applications using Postman, testing response accuracy, performance, and functionality.
- Built reports and data visualizations from backend databases, delivering data-driven insights to stakeholders.

PROJECT

Resume-Aware AI Chatbot

Cloudflare Workers, React, TypeScript, OpenAI API, RAG, Embeddings, KV, R2

- Designed and shipped a serverless RAG chatbot that answers questions about professional background, using document chunking, embeddings, and retrieval pipelines.
- Built production-grade infrastructure, including KV caching, R2 storage, CORS configuration, dataset versioning, and CI/CD deployment.

EDUCATION

Louisiana State University Shreveport, MBA

Jul 2025

Shreveport, LA

- Pursued concurrently with full-time engineering work, with a focus on technology leadership and product strategy.

Mississippi State University, B.S. in Computer Engineering

May 2019

Starkville, MS

- Honors: Undergraduate Research Award, Phi Theta Kappa Scholarship, President's Honor Roll

CERTIFICATIONS & AWARDS

Certifications: Infosys React Professional Certification

Awards: 2x Rising Star Award (Infosys Limited); Undergraduate Research Award (Mississippi State University)